



**Generation Interconnection
System Impact Study Report
for
Queue Project AE1-157
LADYSMITH CT-ST. JOHNS 230 KV
77.8 MW Capacity / 120 MW Energy**

Revision 1 / January 2022

August, 2019

Table of Contents

1	Introduction.....	4
2	Preface.....	4
3	Revision 1 (December 2021) Summary	5
4	General	5
4.1	Point of Interconnection	6
4.2	Cost Summary.....	6
5	Transmission Owner Scope of Work	6
5.1	Attachment Facilities	6
5.2	Direct Connection Cost Estimate.....	7
5.3	Non-Direct Connection Cost Estimate	7
6	Interconnection Customer Requirements.....	8
6.1	System Protection.....	8
6.2	Compliance Issues and Interconnection Customer Requirements	8
6.3	Power Factor Requirements.....	9
7	Revenue Metering and SCADA Requirements	9
7.1	PJM Requirements	9
7.1.1	Meteorological Data Reporting Requirement.....	9
7.2	Dominion Requirements.....	9
8	Network Impacts	9
9	Generation Deliverability	11
10	Multiple Facility Contingency	11
11	Contribution to Previously Identified Overloads	11
12	Potential Congestion due to Local Energy Deliverability.....	11
13	System Reinforcements.....	13
14	Flow Gate Details	14
14.1	Contingency Descriptions.....	14
14.2	Index 1	15
14.3	Index 2	17
15	Affected Systems	19
15.1	Duke Energy Progress.....	19
16	Short Circuit.....	21

17	Stability	23
17.1	Executive Summary	23
Attachment 1		25

1 Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the Feasibility Study Agreement between Urban Grid Solar Projects LLC, the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (VEPCO).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 Revision 1 (January 2022) Summary

This revision is being issued to incorporate results of a re-tool performed in January of 2022 as well as the Stability Analysis and Reactive Power Assessment results.

The reactive power assessment for this project reveals that the project meets the reactive power requirements at the high side of the main transformer.

Additionally, the overdutied breakers previously identified as overdutied have been replaced and are no longer identified as overdutied.

4 General

The IC has proposed a solar and storage generating facility located in Caroline County, Virginia. The installed facilities will have a capability of 120 MW with 77.8 MW of this output being recognized by PJM as Capacity. The proposed in-service date for the AE1-157 project is 12/01/2021. This study does not imply an ITO commitment to either in-service date.

Queue Number	AE1-157
Project Name	LADYSMITH CT-ST. JOHNS 230 KV
Interconnection Customer	Urban Grid Solar Projects LLC
State	Virginia
County	Caroline
Transmission Owner	Dominion
MFO	120
MWE	120
MWC	77.8
Fuel	Solar
Basecase Study Year	2022

4.1 Point of Interconnection

AE1-157 will interconnect with the ITO transmission system via a new three breaker ring bus switching station that connects on the Ladysmith CT – St Johns 230kV line. See one line in Attachment 1.

4.2 Cost Summary

The AE1-157 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 1,800,000
Direct Connection Network Upgrade	\$ 6,500,000
Non Direct Connection Network Upgrades	\$ 800,000
Total Costs	\$ 9,100,000

In addition, the AE1-157 project may be responsible for a contribution to the following costs

Description	Total Cost
System Upgrades	\$ 0

Note: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

5 Transmission Owner Scope of Work

5.1 Attachment Facilities

Generation Substation: Install metering and associated protection equipment. Estimated Cost \$600,000.

Transmission: Construct approximately one span of 230 kV Attachment line between the generation substation and a new AE1-157 Switching Station. The estimated cost for this work is \$1,200,000.

The estimated total cost of the Attachment Facilities is \$1,800,000. It is estimated to take 18-24 months to complete this work. These preliminary cost estimates are based on typical engineering costs. A more detailed engineering cost estimates are normally done when the IC provides an exact site plan location for the

generation substation during the Facility Study phase. The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Generator Substation	\$ 600,000
Transmission	\$ 1,200,000
Total Attachment Facility Costs	\$ 1,800,000

5.2 Direct Connection Cost Estimate

Substation: Establish the new 230 kV AE1-157 Switching Substation (interconnection substation). The estimated cost of this work scope is \$6,500,000. It is estimated to take 24-36 months to complete this work.

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Substation	\$ 6,500,000
Total Direct Connection Facility Costs	\$ 6,500,000

5.3 Non-Direct Connection Cost Estimate

Transmission: Install transmission structure in-line with transmission line to allow the proposed interconnection switching station to be interconnected with the transmission system. Estimated cost is \$800,000 and is estimated to take 24-30 months to complete.

Remote Terminal Work: During the Facilities Study, ITO's System Protection Engineering Department will review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system. The review is based on maintaining system reliability by reviewing ITO's protection requirements with the known transmission system configuration which includes generating facilities in the area. This review may determine that transmission line protection and communication upgrades are required at remote substations.

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Transmission	\$ 800,000
Total Non-Direct Connection Facility Costs	\$ 800,000

6 Interconnection Customer Requirements

6.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion’s “Dominion Energy Electric Transmission Generator Interconnection Requirements” documented in Dominion’s Facility Interconnection Requirements “Exhibit C” located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

6.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC’s GSU transformer(s).
2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

6.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

7 Revenue Metering and SCADA Requirements

7.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

7.1.1 Meteorological Data Reporting Requirement

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

7.2 Dominion Requirements

See Section 3.4.6 “Metering and Telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

8 Network Impacts

The Queue Project AE1-157 was evaluated as a 120.0 MW (Capacity 77.8 MW) injection at Ladysmith-St. Johns 230kV in the Dominion area. Project AE1-157 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-157 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

9 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

10 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
1	314196	6LADYSMITH	DVP	314911	8LADYSMITH	DVP	1	DVP_P1-2: LN 2032-B	single	848	104.42	110.54	AC	50.89
2	939260	AE1-157 TAP	DVP	314197	6LDYSMITH CT	DVP	1	DVP_P1-2: LN 2032-B	single	899	104.61	112.95	AC	77.39

12 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
-	314134	6CRANES	DVP	314142	6STAFORD	DVP	1	DVP_P1-2: LN 2032-B	operation	679	89.89	93.56	AC	25.93
-	314196	6LADYSMITH	DVP	314911	8LADYSMITH	DVP	1	DVP_P1-2: LN 2032-B	operation	848	97.99	107.27	AC	78.49
-	314197	6LDYSMITH CT	DVP	314196	6LADYSMITH	DVP	1	DVP_P1-2: LN 2032-B	operation	984	82.57	90.49	AC	78.49

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
-	314212	6FOUR RIVERS	DVP	939750	AE1-206 TAP	DVP	1	DVP_P1-2: LN 256-B	operation	1123	83.4	93.83	AC	119.36
-	314222	6HANOVER	DVP	314218	6ELMONT	DVP	1	DVP_P1-2: LN 256-B	operation	1123	77.95	88.24	AC	119.36
-	314905	8CHANCE	DVP	314900	8BRISTER	DVP	1	DVP_P1-2: LN 594	operation	2442	124.17	125.22	AC	25.03
-	314908	8ELMONT	DVP	314911	8LADYSMITH	DVP	1	DVP_P1-2: LN 256-B	Operation	2442	102.4	104.1	AC	42.23
-	939260	AE1-157 TAP	DVP	314197	6LDYSMITH CT	DVP	1	DVP_P1-2: LN 2032-B	operation	899	104.61	117.67	AC	119.36
-	939750	AE1-206 TAP	DVP	314222	6HANOVER	DVP	1	DVP_P1-2: LN 256-B	operation	1123	83.39	93.7	AC	119.36
-	939750	AE1-206 TAP	DVP	314222	6HANOVER	DVP	1	Base Case	operation	1061	80.76	84.85	AC	44.98

13 System Reinforcements

ID	Index	Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AE1-157 Allocation
1	1	8LADYSMITH 500/230 kV transformer	PJM baseline upgrade b3027.1: Add a 2nd 500/230 kV 840 MVA transformer at Dominions Ladysmith Substation. The baseline project is in-service date as of 2021.	b3027.1	\$0	\$0
2	2	AE1-157 TAP-6LDYSMITH CT 230 kV line	Four Rivers – St. Johns 230kV line # 256 wave trap at Four Rivers and line switches at St. Johns replaced. Work completed in 2017 and new rating of 876 MVA (normal), 956 MVA (emergency) and 1163 MVA (load dump).	n4692	\$0	\$0
			TOTAL COST		\$0	\$0

14 Flow Gate Details

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gauge other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

14.1 Contingency Descriptions

Contingency Name	Contingency Definition
DVP_P1-2: LN 2032-B	CONTINGENCY 'DVP_P1-2: LN 2032-B' OPEN BRANCH FROM BUS 939750 TO BUS 314222 CKT 1 /* AE1-206 TAP 230.00 - 6HANOVER 230.00 OPEN BRANCH FROM BUS 314218 TO BUS 314222 CKT 1 /* 6ELMONT 230.00 - 6HANOVER 230.00 OPEN BUS 314222 /* ISLAND: 6HANOVER 230.00 END
DVP_P1-2: LN 256-B	CONTINGENCY 'DVP_P1-2: LN 256-B' OPEN BRANCH FROM BUS 939260 TO BUS 314197 CKT 1 /* AE1-157 TAP 230.00 - 6LDYSMITH CT230.00 END
DVP_P1-2: LN 594	CONTINGENCY 'DVP_P1-2: LN 594' OPEN BRANCH FROM BUS 314916 TO BUS 314934 CKT 1 /* 8MORRSVL 500.00 - 8SPOTSYL 500.00 END
Base Case	

14.2 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
1	314196	6LADYSMITH	DVP	314911	8LADYSMITH	DVP	1	DVP_P1-2: LN 2032-B	single	848	104.42	110.54	AC	50.89

Bus #	Bus	MW Impact
315033	1BIRCHWDA	13.58
315043	1FOUR RIVERA	13.55
315044	1FOUR RIVERB	10.5
315045	1FOUR RIVERC	13.55
315046	1FOUR RIVERD	10.5
315047	1FOUR RIVERE	10.5
315048	1FOUR RIVERF	13.54
315037	1LDYSMT1	17.41
315038	1LDYSMT2	17.4
315039	1LDYSMT3	18.41
315040	1LDYSMT4	18.45
315041	1LDYSMT5	18.5
315034	1NORNECKC1	0.56
315035	1NORNECKC2	0.54
315005	1POSSM 3	12.15
314131	6ARNOLDS	0.48
314134	6CRANES	0.11
933011	AC2-125	9.75
933021	AC2-126	9.81
933271	AC2-138 C	0.11
936341	AD2-044 C	0.19
936581	AD2-073 C	2.08
939261	AE1-157 C O1	50.89
939271	AE1-158 C O1	51.93
939751	AE1-206 C O1	117.73
LTF	BLUEG	5.16
LTF	CALDERWOOD	0.63
LTF	CANNELTON	0.32
LTF	CARR	0.14
LTF	CATAWBA	0.44
LTF	CHEOAH	0.58
LTF	COFFEEN	0.55
LTF	COTTONWOOD	2.33
LTF	DUCKCREEK	1.18
LTF	EDWARDS	0.54
LTF	FARMERCITY	0.37
LTF	GIBSON	0.21
LTF	HAMLET	0.74
LTF	NEWTON	1.43
LTF	PRAIRIE	2.73
LTF	RENSSELAER	0.11

Bus #	Bus	MW Impact
LTF	SMITHLAND	0.22
LTF	TATANKA	0.65
LTF	TILTON	0.64
LTF	TRIMBLE	0.57
LTF	TVA	1.95
LTF	UNIONPOWER	0.88
919211	AA1-145	39.17
924061	AB2-050	2.3
926471	AC1-118 C	0.41
926551	AC1-134	5.76

14.3 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
2	939260	AE1-157 TAP	DVP	314197	6LDYSMITH CT	DVP	1	DVP_P1-2: LN 2032-B	single	899	104.61	112.95	AC	77.39

Bus #	Bus	MW Impact
315043	1FOUR RIVERA	20.61
315044	1FOUR RIVERB	15.96
315045	1FOUR RIVERC	20.61
315046	1FOUR RIVERD	15.96
315047	1FOUR RIVERE	15.96
315048	1FOUR RIVERF	20.59
939261	AE1-157 C O1	77.39
939271	AE1-158 C O1	78.98
939751	AE1-206 C O1	179.04
LTF	BLUEG	2.71
LTF	CALDERWOOD	0.29
LTF	CANNELTON	0.16
LTF	CARR	0.2
LTF	CATAWBA	0.19
LTF	CHEOAH	0.27
LTF	COFFEEN	0.29
LTF	COTTONWOOD	1.13
LTF	DUCKCREEK	0.62
LTF	EDWARDS	0.28
LTF	FARMERCITY	0.19
LTF	GIBSON	0.11
LTF	HAMLET	0.31
LTF	NEWTON	0.75
LTF	PRAIRIE	1.39
LTF	RENSSELAER	0.16
LTF	SMITHLAND	0.11
LTF	TATANKA	0.34
LTF	TILTON	0.34
LTF	TRIMBLE	0.3
LTF	TVA	0.95
LTF	UNIONPOWER	0.42
919211	AA1-145	59.58
924061	AB2-050	3.5
926551	AC1-134	8.76

Affected Systems

15 Affected Systems

15.1 Duke Energy Progress

None

Short Circuit

16 Short Circuit

The following Breakers are overdutied:

None

Stability

17 Stability

17.1 Executive Summary

Generator Interconnection Request AE1-157 is for a 120.0 MW Maximum Facility Output (MFO) solar generating facility. AE1-157 consists of 46 x TMEIC PVH-L2700GR inverters rated at 2.67 MW each. The Point of Interconnection (POI) is at a tap on St. Johns – Ladysmith CT 230 kV circuit in the Dominion Virginia Power (DVP) system, Caroline County, Virginia.

Generator Interconnection Request AE1-158 is for a 120.0 MW Maximum Facility Output (MFO) solar generating facility. AE1-158 consists of 48 x TMEIC PVH-L2700GR inverters rated at 2.61 MW each. AE1-158 shares the POI with AE1-157 at a tap on St. Johns – Ladysmith CT 230 kV circuit in the Dominion Virginia Power (DVP) system, Caroline County, Virginia.

This report describes a dynamic simulation analysis of AE1-157 and AE1-158 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2022 peak load case, modified to include applicable queue projects. AE1-157 and AE1-158 have been dispatched online at maximum power output, with 1.0 p.u. voltage at the generator bus.

AE1-157 and AE1-158 were tested for compliance with NERC, PJM, Transmission Owner and other applicable criteria. Steady-state condition and 53 contingencies were studied, each with a 20 second simulation time period. Studied faults included:

- a) Steady-state operation (30 second);
- b) Three-phase faults with normal clearing time;
- c) Single-phase faults with stuck breaker;
- d) Single-phase faults placed at 80% of the line with delayed (Zone 2) clearing at line end remote from the fault due to primary communications/relay failure;
- e) Single phase faults with loss of multi-circuit tower line.

No relevant bus fault and high-speed reclosing (HSR) fault were identified for this study.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For all of the fault contingencies tested on the 2022 peak load case:

- a) AE1-157 and AE1-158 were able to ride through the faults (except for faults where protective action trips a generator(s)),
- b) The system with AE1-157 and AE1-158 included is transiently stable and post-contingency oscillations were positively damped with a damping margin of at least 3% for interarea modes and 4% for local modes.

- c) Following fault clearing, all bus voltages recovered to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element tripped, other than those either directly connected or designed to trip as a consequence of that fault.
- e) North Anna 500 kV bus voltage is within 510 kV – 535.5 kV range with less than 3.5% voltage drop for all contingencies, per PJM M03.

The reactive power capability of AE1-157 meets the 0.95 lagging and leading PF requirement at the high side of the main transformer.

The reactive power capability of AE1-158 meets the 0.95 lagging and leading PF requirement at the high side of the main transformer.

Tripping of Ladysmith and Four Rivers units was observed for contingencies P4.14, P4.21 and P4.22 which were resolved by using the actual fault impedance calculated from the short circuit case.

Slow QELEC response of AE1-157 and AE1-158 was observed in P1.11. However, the POI voltage was within acceptable bandwidth and slow QELEC response did not cause any instability in the system. Additional tuning can be performed if required.

No mitigations were found to be required.

Attachment 1

System Configuration

